Conflict-Complete Retrieval: Checkable Contradictions under a Context Budget

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Abstract

An agent cannot act on a contradiction that memory retrieval does not expose. We define *conflict completeness*: for every query-relevant conflict, a retrieval policy returns at least one complete, source-grounded witness that a sound checker can verify from the context alone. We characterize the exact budget-feasibility boundary by the minimum joint union of one witness per conflict. A capped exact selector realizes this boundary on tractable classes and otherwise abstains rather than silently returning incomplete evidence. We also show that every fixed-hop policy based only on IRI adjacency fails conflict completeness on a literal-keyed construction. A Conflict Witness Index (CWI) incrementally maintains key-induced record equivalence and returns one shortest witness per conflict. Evaluation covers 252 synthetic instances with 132 conflicts, 12 multi-conflict queries, and 136 semi-natural controls on gold record-linkage topology. Shared premises reduce a 32-conflict joint certificate from 192 to 68 triples, whereas 32 disjoint conflicts require all 192; on an alternative-path case, exact joint selection reduces 33 triples to the proven optimum 27. Exact maintained, exact on-demand, and adaptive exact retrieval return the same 2–12 triples per individual conflict; learned baselines require up to whole 336–440-triple stores on key chains. All semantic conflict labels are controlled and the schema is supplied: this is a retrieval contract and cost study, not evidence of natural conflict prevalence.

Introduction

Long-lived agents increasingly move information between a large external memory and a bounded language-model context. Systems such as MemGPT explicitly manage memory tiers (Packer et al. 2023), while graph retrieval systems such as HippoRAG integrate information across stored passages (Gutiérrez et al. 2024). These systems are normally evaluated on whether retrieval supports an answer. We ask a different question: if the store already contains incompatible answers, what must retrieval return so the conflict is not silently hidden?

Consider two records that are declared to identify the same person through an email key, but give different values for a single-valued predicate. A consumer shown only one record cannot distinguish "no conflict exists" from "the other premise was not retrieved." Returning a Boolean flag is also insufficient for audit: the consumer cannot check the sources. Returning the whole store is complete but defeats bounded retrieval.

Our first contribution is a contract between storage and a downstream reasoner: for each relevant conflict, return one whole minimal witness. Minimal explanations themselves are classical justifications, conflict sets, and why-provenance (Reiter 1987; Kalyanpur et al. 2007; Baader and Peñaloza 2010; Buneman, Khanna, and Tan 2001; Green, Karvounarakis, and Tannen 2007). We do not claim a new conflict detector or witness formalism. What is new is to quantify checkable-witness inclusion over the *returned context*, characterize when all relevant conflicts can fit a fixed budget, and require explicit overflow otherwise. Second, we prove a boundary for IRI-only neighborhoods. Third, we implement both fast independent selection and bounded exact joint selection, and measure maintained retrieval against exact, key-aware, and learned baselines.

The experiments produce five findings. First, a key-aware neighborhood is a strong baseline: for direct conflicts it is usually within a few triples of an exact witness. Second, chained identity is the regime where record-level neighborhood and relevance retrieval become expensive. Third, a maintained index and a fresh exact join return the same witness-sized output, but a fair ingest-plus-query workload identifies the maintenance break-even. Fourth, joint context cost depends on premise sharing, and exact joint selection can avoid false overflow by choosing longer but more shareable witnesses. Fifth, the mechanism transfers to controlled conflicts over independently supplied record-linkage topology and natural distractor attributes. These boundaries are central to the claim.

Conflict-Complete Retrieval

Store and schema. An assertion is $a = (s, p, o, \tau, \sigma, \pi)$: an RDF-like triple, a valid-time interval, a scope, and provenance. A confidence threshold θ defines the premise store S_θ. Missing confidence is excluded. Literals use exact term identity; no normalization or entity linker is assumed.

The oracle schema declares a single key predicate k_C for a keyed class C and a set E of predicates that are single-valued per entity, overlapping time, and scope. Records x and y are directly co-keyed, written $x \approx_K y$, when both have type C and the same k_C value. Let $\sim_K$ be its reflexive, symmetric, transitive closure. Bottom time or scope overlaps everything; otherwise time intervals must intersect and scopes must be equal.

A conflict $\gamma = (e, p, \{v_1, v_2\}, \tau, \sigma)$ exists when $p \in E$, $v_1 \neq v_2$, and two records in $[e]_{\sim_K}$ assert the values over overlapping time and scope. This preserve-and-surface semantics differs from repair-based consistent query answering (Arenas, Bertossi, and Chomicki 1999): both assertions remain in the store and the violation becomes an explicit object.

Rule fragment. The executable semantics has three rules. First, the two type and two matching key assertions derive $x \sim y$. Second, equivalence propagates values only for predicates in E; it is not unrestricted `sameAs` replacement. Third, two unequal E-values derive a conflict exactly when their annotations overlap. The least fixpoint is finite: key equivalence partitions finitely many records, propagation copies only finitely many stored E-values across a partition, and violation generation is pairwise within a partition. The confidence gate filters leaves before any rule fires.

Definition 1 (Witness)**.** *A set $W \subseteq S_\theta$ is a witness for conflict γ if the key, type, value, time, and scope premises in W entail γ, and no proper subset does. Let $\mathcal{W}_\gamma$ be all such witnesses.*

A same-record witness has two value assertions. A direct cross-record witness has two types, two matching keys, and two conflicting values, hence six logical triples. For a shortest chain of m co-key links, intermediate types are shared across adjacent links, giving

$$|W| = (m+1) + 2m + 2 = 3m + 3. \tag{1}$$

Serialized certificates additionally carry any record-level time/scope metadata required to reproduce the annotated assertions.

Proposition 1 (Detector soundness and completeness)**.** *For the rule fragment above, the least fixpoint derives a conflict if and only if the quotient store under $\sim_K$ contains two unequal values for a predicate in E with overlapping time and scope. Every minimal set of source leaves in such a derivation is a witness.*

Proof. For soundness, key-rule premises satisfy direct co-keying, so propagation moves an E-value only between representatives of the same quotient entity. The violation rule then has exactly the two unequal, overlap-compatible facts required by the conflict semantics. For completeness, take any semantic conflict. Its endpoint records lie in one finite $\sim_K$ class and therefore have a path of direct co-key steps. Repeated propagation carries both endpoint values to one representative, where the violation rule fires. Removing redundant derivation leaves until inclusion-minimal gives a witness. □

A query q has entity seeds $\mathrm{ent}(q)$ and predicates $\mathrm{pred}(q)$. A conflict is relevant if its equivalence class intersects the seeds and its predicate is requested. Write $\Gamma(S_\theta, q)$ for the relevant conflicts.

Definition 2 (Conflict completeness)**.** *A policy $\mathcal{R}$ is conflict-complete for (S, q) iff*

$$\forall \gamma \in \Gamma(S_\theta, q)\ \exists W \in \mathcal{W}_\gamma : W \subseteq \mathcal{R}(q, S). \tag{2}$$

It is (B, μ)-conflict-complete when Eq. 2 holds and $\mu(\mathcal{R}(q, S)) \leq B$ for a context-cost measure μ.

Assume μ is monotone under set inclusion. The least possible joint certificate cost is

$$b^\star(S, q) = \min_{\{W_\gamma \in \mathcal{W}_\gamma\}_{\gamma \in \Gamma}} \mu\left(\bigcup_{\gamma \in \Gamma} W_\gamma\right), \tag{3}$$

with $b^\star = 0$ when Γ is empty.

Proposition 2 (Budget feasibility)**.** *A (B, μ)-conflict-complete result exists for (S, q) if and only if $b^\star(S, q) \leq B$.*

Proof. If the minimum union in Eq. 3 fits, returning that union satisfies Eq. 2. Conversely, every complete result contains some witness for each relevant conflict; monotonicity makes its cost at least $b^\star$. □

Proposition 3 (Joint selection complexity)**.** *Given explicit finite witness families and triple-count cost, deciding whether $b^\star(S, q) \leq B$ is NP-complete.*

Proof. Membership is immediate. Reduce Set Cover: create a conflict γ_e for each universe element e, a shared assertion token a_j for each candidate set A_j, and witness options $\{d_e, a_j\}$ whenever $e \in A_j$, where d_e is conflict-specific. A joint union costs $|\Gamma|$ plus the number of selected set tokens, so a budget exists exactly when a cover of the corresponding size exists. □

Equation 2 requires one complete proof per conflict, not every alternative proof. We call the latter, stronger property *proof-exhaustive completeness*. The distinction matters because a conflict may have exponentially many minimal proofs, while one is sufficient for checking. Proposition 2 also exposes an unavoidable boundary: if $B < b^\star$, no retrieval algorithm can return complete evidence for every relevant conflict. It must enlarge the context, narrow the requested predicate/seed footprint, or report overflow.

What the contract does and does not guarantee. Conflict completeness is a property of the returned assertions, independent of the checker or language model that consumes them. It prevents retrieval from being the reason a finite, relevant conflict is unverifiable. It does not guarantee that an unsound consumer notices the conflict, that the oracle schema is correct, or that every real inconsistency has a finite witness in this fragment. Conversely, a policy can satisfy ordinary answer relevance while failing Eq. 2; seeing one plausible value is insufficient.

A Boundary for IRI-Only Neighborhoods

Let $G_I(S)$ contain an undirected edge only when two IRIs co-occur as subject and object in a triple. Literals are not graph nodes. A k-hop IRI policy may return assertions incident to IRIs within distance k of a query seed.

Proposition 4 (Fixed-hop incompleteness)**.** *For every finite k, there is a store and query with a relevant six-assertion cross-record witness for which any policy restricted to the k-hop component of $G_I(S)$ is not conflict-complete.*

Proof. Create typed records x and y with the same *literal* key u, and assert different values of $p \in E$. Seed the query at x. Add no IRI-object path from x to y. Then $x \sim_K y$, so the six assertions form a relevant witness. But y is in a different component of $G_I(S)$ and its type, key, and value assertions cannot be returned at any finite k. □

This result is deliberately narrow. It does not apply to literal-aware, key-aware, lexical, dense, or hybrid policies. On connected benchmark stores, IRI breadth-first search can also become complete by expanding to most or all of the store. The proposition establishes a counterexample class, not a claim about every graph-retrieval workload.

Conflict Witness Index

CWI maintains: (i) buckets indexed by class, key predicate, and exact key value; (ii) union–find classes induced by bucket membership; (iii) single-valued assertions and time/scope annotations per class; and (iv) a record–bucket graph for certificate reconstruction. A touching insertion or metadata update re-evaluates the affected equivalence class. At query time, CWI selects one shortest record–bucket path for each relevant conflict and returns the corresponding source assertions.

Proposition 5 (CWI contract)**.** *Under the oracle schema and exact-literal semantics, after processing a store snapshot, CWI is conflict-complete and each returned certificate is a minimal witness. A conflict whose shortest supporting chain has length m uses $3m+3$ logical assertions, or two for a same-record conflict. Let $\widehat{U}$ be the union selected independently across $g = |\Gamma|$ conflicts. For triple count, $|\widehat{U}| \leq g\, b^{\star}$. Budget-aware CWI returns $\widehat{U}$ when $|\widehat{U}| \leq B$ and otherwise returns* SELECTION-OVERFLOW *with no partial context.*

Proof sketch. Union–find contains exactly the transitive closure of direct co-key buckets. Re-evaluating every constrained predicate in a touched class enumerates exactly pairs with unequal values and overlapping time/scope. A shortest record–bucket path supplies all and only the type and key premises connecting the endpoint value assertions. Removing an endpoint value breaks the violation; removing a path premise breaks co-reference. Thus the assembled set is minimal and satisfies Eq. 2. Each selected shortest witness is no larger than the corresponding witness in an optimal union, which is itself no larger than that union; summing over g gives the stated bound. □

Overflow does not prove that $B < b^{\star}$: alternative witnesses could share more premises than CWI's independently selected shortest paths. It is an explicit refusal to mislabel a truncated context as complete. CWI is an incremental materialized view (Gupta, Mumick, and Subrahmanian 1993), specialized to retrieval of explanations.

Proposition 6 (Exact joint CWI)**.** *In this rule fragment, enumerate every simple record–bucket path supporting each relevant conflict and apply exact branch-and-bound selection to the resulting witness families. If both stages finish exhaustively, the returned union has cost $b^{\star}$; the method reports* INFEASIBLE *exactly when $B < b^{\star}$. A capped run may instead report* SEARCH-LIMIT*, with no partial context.*

Proof sketch. After removing cycles and unused premises, every minimal cross-record identity proof is a simple path; endpoint values and required annotations complete its witness. Thus exhaustive path enumeration yields $\mathcal{W}_\gamma$ for every γ. Exhaustive branch-and-bound considers every choice in Eq. 3, so its best union is $b^{\star}$. A found fitting union is always safe; only exhaustive failure justifies INFEASIBLE. □

Update and query path. On insertion, CWI records a type, key, constrained value, or annotation. A record enters a key bucket only after both its class and key are known. Joining a non-singleton bucket unions its members and merges their constrained value lists. Every affected predicate cell is then re-evaluated, allowing a later time/scope annotation to retract or restore a conflict in the current record snapshot. Querying a seed locates its union–find root, filters conflicts by the query predicate footprint, and runs breadth-first search on the record–bucket graph only to reconstruct a shortest source path. Output is sorted by stable triple identifier.

Complexity. Let n_c be records in an affected equivalence class and r_p its assertions for constrained predicate p. Bucket insertion has inverse-Ackermann union–find overhead; merging and re-evaluating a touched class is $O(n_c + \sum_p r_p^2)$ in the current pair-enumerating implementation. This quadratic class-local term is a genuine worst case, hidden by the bounded-class generator used below. Index storage is linear in source types, keys, constrained values, and the materialized conflicting pairs. A query is linear in the conflicts returned plus the record–bucket edges visited for their shortest paths. These bounds motivate reporting update amplification and class-controlled scaling rather than claiming store-size-independent maintenance. Alternative-path enumeration and joint selection are exponential in the worst case, as Proposition 6 and the NP-completeness result require; explicit path and search-node caps preserve safe abstention.

Evaluation

Questions. We test: (Q1) which policies return a complete witness and at what context cost; (Q2) whether maintained CWI improves certificate size over exact on-demand retrieval; (Q3) where retained maintenance breaks even against fresh exact queries; (Q4) how joint selection behaves with shared, disjoint, and alternative witnesses; (Q5) how the disclosed class-local quadratic term manifests; and (Q6) whether the mechanism transfers to natural linkage topology. Labels are deterministic and the schema is supplied as ground truth. This isolates retrieval but cannot measure schema induction, extraction errors, or natural prevalence.

Fixtures. Eight families contain 252 query instances and 132 conflicts. D20 exercises same-record and direct-key disagreement; XR-small and XR-scale add distracting records; Chain-m requires $m \in \{2, 3\}$ co-key links; H3 varies hierarchy and non-key predicates; and TauSig adds record-snapshot time/scope metadata. Benign controls include shared non-key values, distinct entities, disjoint time, and unequal scope. Chain lengths $m = 0, 1, 2, 3$ are exercised. Separately, twelve deterministic multi-conflict queries vary $g \in \{1, 2, 4, 8, 16, 32\}$ with either one shared identity spine or g disjoint classes. These unique-path constructions make $b^\star$ exactly known. An additional four-conflict graph provides 20 alternative minimal witnesses, including longer paths whose joint union is smaller. The semi-natural leg uses component-disjoint pairs from a public entity-resolution benchmark.

Systems. CWI is compared with a fresh full-store exact key/equivalence join per query (*on-demand exact*) and an *adaptive exact* baseline that expands key-aware layers and stops once it can verify a positive certificate. Adaptive search does not generally certify absence at its configured eight-layer cap. Record-level policies retrieve all assertions of selected records: IRI BFS and key-aware neighborhoods at depths 1–4; BM25 (Robertson and Zaragoza 2009); cosine retrieval with `all-MiniLM-L6-v2` at a pinned model revision (Reimers and Gurevych 2019); and BM25–dense reciprocal-rank fusion with $k_{\mathrm{RRF}} = 60$ (Cormack, Clarke, and Buettcher 2009). Relevance methods use top-$K \in \{1, 2, 4, \ldots, 512\}$.

Protocol and implementation. TypeScript systems run under Node.js 22.17.1; the dense baseline uses Sentence Transformers 5.6.1. The encoder weights, exact revision, and Python dependency lock are included in the artifact. Scaling ran on one AMD Ryzen Threadripper PRO 5955WX host with 256 GiB RAM and AlmaLinux 9.7; no GPU was used. Each fixture, schema, output row, and scale store has a deterministic identifier or SHA-256 digest. Retrieval configurations run over identical serialized stores; the exact checker then evaluates the context returned by each policy. No hosted model call contributes to a confirmatory result. Scale timings use one unrecorded warm-up followed by 11 measured repetitions; we report medians and interquartile ranges (IQR). The large-class stress leg uses the same repetition protocol. The maintenance workload materializes the same source triples in memory before both arms: CWI pays one insertion batch plus q lookups, while the stateless arm rebuilds its exact index and closure for each query.

We score a conflict only when the retrieved context contains a complete minimal witness. Context cost is serialized triple count; Table 1 reports the smallest configuration that reaches 100% conflict completeness within a family. Values are mean/maximum over conflict queries. Since fixtures and outputs are deterministic, we report exact counts rather than significance tests.

Retrieval Results

Exactness check. Maintained CWI and the full exact equivalence join both produced 132 true positives, 120 true negatives, zero false positives, and zero false negatives. Every emitted witness was rechecked from its returned triples without dereferencing the store. In TauSig, the twelve positive cases include the required metadata premises; 48 controls remain benign because their time or scope cells do not overlap. This is a conformance check for the implementation and fixtures, not evidence that the supplied schema reflects a real domain.

The strongest simple baseline is strong. Key-aware depth one returns every direct conflict in 4–12 triples. Thus, when identity is direct, a maintained witness index saves little context. XR-scale also shows that BM25 can be ideal when the exact key string is present in the query: its eight triples match key-aware retrieval despite a 1,689-triple store. Dense retrieval is not uniformly competitive on exact identifier matching.

Chains expose record-level expansion. For Chain-2 and Chain-3, exact witnesses contain 9 and 12 triples as predicted by Eq. 1. Key-aware retrieval must include the radius ball and averages 94.6 and 91.3 triples. All three relevance methods reach full completeness only at their respective whole-store contexts (440 and 336 triples). This does not show that learned retrieval cannot be trained for the task; it shows that standard relevance ranking provides no completeness contract on these controlled chains.

Maintenance does not shrink the certificate. CWI, on-demand exact, and adaptive exact return the same shortest witness on every conflict. Therefore CWI has no output-size advantage over fresh exact computation. Adaptive expansion also finds witness-sized positive answers without maintained state, though it scans increasing layers and its finite cap does not prove benign absence. The operational choice is between write-time maintenance, query-time exact work, and absence guarantees.

Joint Budgets and Explicit Overflow

Table 2 shows both sides of the budget boundary. The shared family reuses four identity premises, so its exact joint cost is $4 + 2g$ rather than $6g$; the disjoint family shares nothing and requires $6g$. Across all 12 queries, CWI returns every conflict at $B = b^\star$ and returns SELECTION-OVERFLOW, with an empty context, at $B = b^\star - 1$. This is a contract conformance study. In the alternative-path case, independent shortest selection needs 33 triples. Exact joint CWI explores 20 candidate witnesses (226 search nodes), proves a 27-triple optimum, succeeds at $B = 27$ where the independent policy overflows, and proves infeasibility at $B = 26$.

Semi-Natural External Evaluation

We use the CC-BY DBLP–Google Scholar entity-resolution benchmark (Papadakis 2022). Gold match edges and their transitive topology are independently supplied; title, author, venue, and year fields remain as natural distractors. To avoid relabeling lexical differences as semantic contradictions, we inject one controlled single-valued predicate at 68 component-disjoint record pairs: unequal endpoint values form positives and equal values form paired negatives. The

Family	Instances/conflicts	Store	Exact witness	Key-aware	BM25	Dense	Hybrid
D20	14/8	133	2.0/2	4.0/4	7.5/8	7.8/8	7.4/8
XR-small	16/12	345	6.0/6	8.0/8	83.0/83	345.0/345	163.0/163
XR-scale	64/60	1,689	6.0/6	8.0/8	8.0/8	1689.0/1689	323.0/323
Chain-2	41/10	440	9.0/9	94.6/98	440.0/440	440.0/440	440.0/440
Chain-3	25/6	336	12.0/12	91.3/112	336.0/336	336.0/336	336.0/336
H3-NK1	16/12	405	6.0/6	9.0/9	99.0/99	405.0/405	195.0/195
H3-NK3	16/12	525	6.0/6	11.0/11	244.8/254	525.0/525	493.3/495
TauSig	60/12	648	10.0/10	12.0/12	220.2/226	648.0/648	430.8/433

Table 1: Triples returned at the smallest fully conflict-complete setting (mean/maximum). "Exact witness" is identical for maintained CWI, on-demand exact, and positive adaptive exact output.

g	Shared sum	Shared union	Disjoint union
1	6	6	6
2	12	8	12
4	24	12	24
8	48	20	48
16	96	36	96
32	192	68	192

Table 2: Multi-conflict triple budgets. "Sum" adds certificates without deduplication; unions deduplicate shared premises and equal $b^\star$ in these unique-path constructions.

resulting 136 instances contain 20 positive pairs at each path depth 1–3 and eight at depth 4.

CWI is correct on all 68 positives and 68 negatives. Natural component contexts span 12–224 triples (median 32), including 6–113 distractor triples; returned positive witnesses remain 6–15 triples as predicted by Eq. 1. This establishes transfer to real linkage topology and record attributes, not natural conflict prevalence: the conflict predicate and schema remain controlled.

Maintenance Workload and Scaling

Table 3 compares one in-memory insertion batch followed by $q \in \{1, 10, 100\}$ queries. Parsing is outside both timed regions. The stateless arm rebuilds the full exact index and equivalence closure per query; CWI retains derived state.

CWI has lower median total time from the first tested query, by 1.56–2.05×, and by 93–186× at 100 queries. It retains 0.820–0.821 retained items per source triple; the fresh arm trades this space for repeated computation. A separate ledger scales CWI through 100,000 persons (560,009 triples): 100 lookups take 0.67 ms median after 672.2 ms ingestion. Conversely, when 256 records share one key and distinct values, CWI materializes all 32,640 conflict pairs and ingestion reaches 253.1 ms median, exposing the quadratic class-local regime.

Related Work

Justifications and axiom pinpointing identify minimal premises for a fixed entailment after access to the knowledge base (Kalyanpur et al. 2007; Baader and Peñaloza 2010); diagnosis uses minimal conflict sets (Reiter 1987); and database provenance tracks which source tuples support an output (Buneman, Khanna, and Tan 2001; Green, Karvounarakis, and Tannen 2007). We reuse these objects and incremental view maintenance (Gupta, Mumick, and Subrahmanian 1993). The distinction is the quantified object: our property ranges over every relevant conflict and the source assertions *returned* under a context budget. Equations 2–3 turn omission into a retrieval failure and separate algorithmic overflow from information-theoretic infeasibility. Unlike per-entailment justification retrieval, exact joint CWI selects across all relevant conflicts under one context budget. This is a contract-and-selection contribution, not a new inconsistency logic.

Long-term LLM memory work studies tiered context management and multi-hop retrieval (Packer et al. 2023; Gutiérrez et al. 2024). Sparse and dense retrieval optimize relevance rather than worst-case witness inclusion (Robertson and Zaragoza 2009; Reimers and Gurevych 2019). Reciprocal rank fusion combines rankings without creating a logical completeness guarantee (Cormack, Clarke, and Buettcher 2009). These objectives are complementary: relevance can select candidate entities, after which conflict-complete retrieval can attach checkable exception evidence.

Limitations and Validity

The strongest limitation is external semantic validity. The 252 core instances are synthetic, while the 136 semi-natural instances preserve real linkage topology and attributes but inject the conflict predicate; all use a supplied schema. A post-audit of exploratory public data found that an earlier vulnerability analysis had incorrectly treated a non-transitive "related" field as identity, while a clinical-trial probe had treated shared grant identifiers as study identity. We removed both prevalence claims rather than substitute a weaker number. The corrected vulnerability alias graph contained no key chains and lacked semantic conflict labels. We therefore make no claim that naturally occurring conflicts or chain-conflict regimes are common in agent memory.

The formal fragment has one exact key per class, exact literal comparison, single-valued constraints, interval overlap, and equal-or-unspecified scope. It excludes learned entity resolution, composite keys, uncertainty propagation, n-ary constraints, and open-world conflicts without finite witnesses. Confidence and schema are trusted inputs. Record-snapshot time/scope metadata is implemented, but arbitrary

Persons	Triples	Entries	$q = 1$ total ms		$q = 100$ total ms	
			CWI	Fresh	CWI	Fresh
300	1,689	1,386	0.69	1.07	1.19	111.26
3,000	16,809	13,806	9.77	20.02	10.39	1,732.23
10,000	56,009	46,006	39.79	75.19	40.31	7,505.44

Table 3: Median total workload time over 11 runs after one warm-up. Both arms receive the same parsed source snapshot. "Entries" includes structural index entries and materialized conflict pairs; the fresh arm retains none. No run had a missing or spurious conflict.

event histories are not.

The index is insertion-oriented; deletion, persistence, concurrent updates, and crash recovery remain engineering work. Independent selection may still return SELECTION-OVERFLOW; exact joint selection closes that gap only when its exponential path and search caps are not reached. The workload is a single-process insertion batch followed by reads; it does not establish concurrent latency, heap residency, or behavior under deletions. The learned baselines use one frozen encoder without task-specific training. Finally, a complete witness guarantees that a sound checker *can* verify a conflict, not that a language model will notice it. Our exploratory reader runs confounded witness size with distractors and used an outcome-defined competence subgroup, so we exclude them from confirmatory evidence.

Generative-AI disclosure. Generative AI tools assisted with language editing, software and proof review, claim-falsification checks, and submission-compliance auditing. The authors verified all text, proofs, citations, code, and results, take responsibility for the complete submission, and list no AI system as an author or source.

Conclusion

Conflict completeness makes a missing contradiction a falsifiable retrieval failure: every relevant conflict must arrive with one checkable witness, and the minimum joint witness union states exactly when that demand can fit. Fixed-hop IRI neighborhoods cannot guarantee it, while exact maintained and on-demand methods can return witness-sized contexts; bounded exact joint selection can additionally prove feasibility or infeasibility on tractable classes. Simple key-aware retrieval is already competitive for direct conflicts, while maintained retrieval wins the tested read workloads at the cost of retained state and quadratic worst-case classes. Independently validated natural schemas and conflict labels remain necessary to establish prevalence in deployed agent memories.

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